Three OTA Styles for a 50/60 Hz Notch in Sky130

Overall comparison: current cancellation, bulk-driven and current division

All three styles were built into the same 3rd-order elliptic OTA-C notch, with the same Sky130 MiM capacitors ($C_2 = C_{L2} = 4.08$ pF, $C_1 = C_3 = 0.52$ pF), the same unity-gain op-amp buffer, the same ±0.9 V supply and the same testbench and analyses. Each style was tuned to 50 Hz and 60 Hz by its own bias current only. So the differences below come from the OTA alone. Detailed derivations are in each style's own report.

1. Side-by-side metrics

The best value in each row is in bold. Monte Carlo yield uses one strict rule for all three: a run counts only if its notch lies between 20 and 200 Hz, is at least 20 dB deep, and its 10 Hz gain is within 6 dB of nominal.

Metric	Current cancellation	Bulk-driven	Current division
Notch depth at 50 Hz, schematic (dB)	**43.1**	38.2	22.5
Notch depth at 60 Hz, schematic (dB)	**41.6**	38.1	25.1
Notch depth at 50 Hz, post-layout (dB)	**41.7**	34.3	21.0
Monte Carlo yield, schematic (of 100)	93	**98**	47
Monte Carlo yield, post-layout (of 100)	94	**100**	33
Monte Carlo f_0 spread, working runs (Hz, σ)	**1.78**	3.15	10.06
f_0 range over corners at fixed bias (Hz)	45.3–54.1	44.9–55.1	47.9–60.3
Gain at 10 Hz (dB)	-7.06	-4.00	-6.59
Gain at 100 Hz (dB)	-12.70	**-6.62**	-10.83
THD at 5 mV input (%)	**0.18**	0.81	0.69
THD at 20 mV input (%)	**0.66**	3.22	2.67
OTA linear range, ±10% g_m (mV)	-23 to +25	-96 to +95	-31 to +33
OTA systematic offset (mV)	**7.6**	18.1	26.1
OTA open-loop gain (dB)	**37.7**	32.3	25.5
OTA CMRR (dB)	55.9	**72.3**	46.2
OTA input noise, 0.05–300 Hz (µV rms)	**150**	332	334
Filter power, 5 OTAs + bias (nW)	**2.32**	4.10	12.33
Bias current for 50 Hz (nA)	0.2096	0.3382	0.0897
One OTA layout (µm²)	4019	**2471**	6960
Step settling to ±2% (ms)	43.8	**40.8**	41.3
Group delay at 10 Hz (ms)	3.64	**2.74**	3.05
Design cycles needed	4	1	4

Shared by all three: the buffer draws 1.565 µW, which is 99% or more of the total power in every case, and the four MiM capacitors take about 4562 µm².

2. What the overlaid plots show

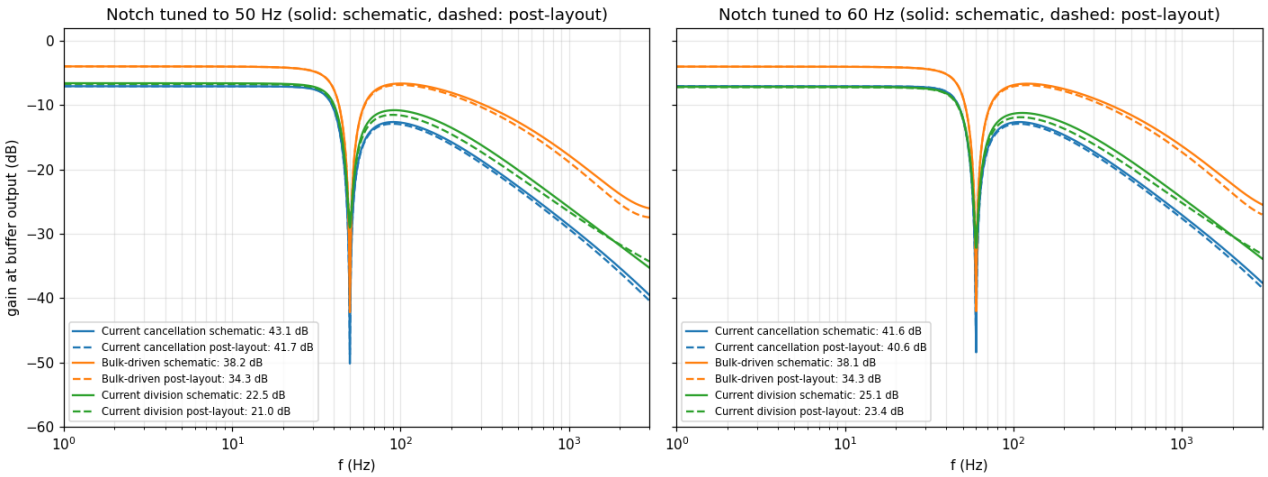


Figure 1. Notch response at the buffer output, tuned to 50 Hz (left) and 60 Hz (right). Solid lines: schematic; dashed: extracted OTA layout.

All three hit 50 and 60 Hz with the same capacitors. Current cancellation gives the deepest notch and loses only about 1 dB to layout. Bulk-driven loses about 4 dB to layout, because each input drives its own n-well and that well's capacitance lands on the filter nodes. Current division is the shallowest throughout.

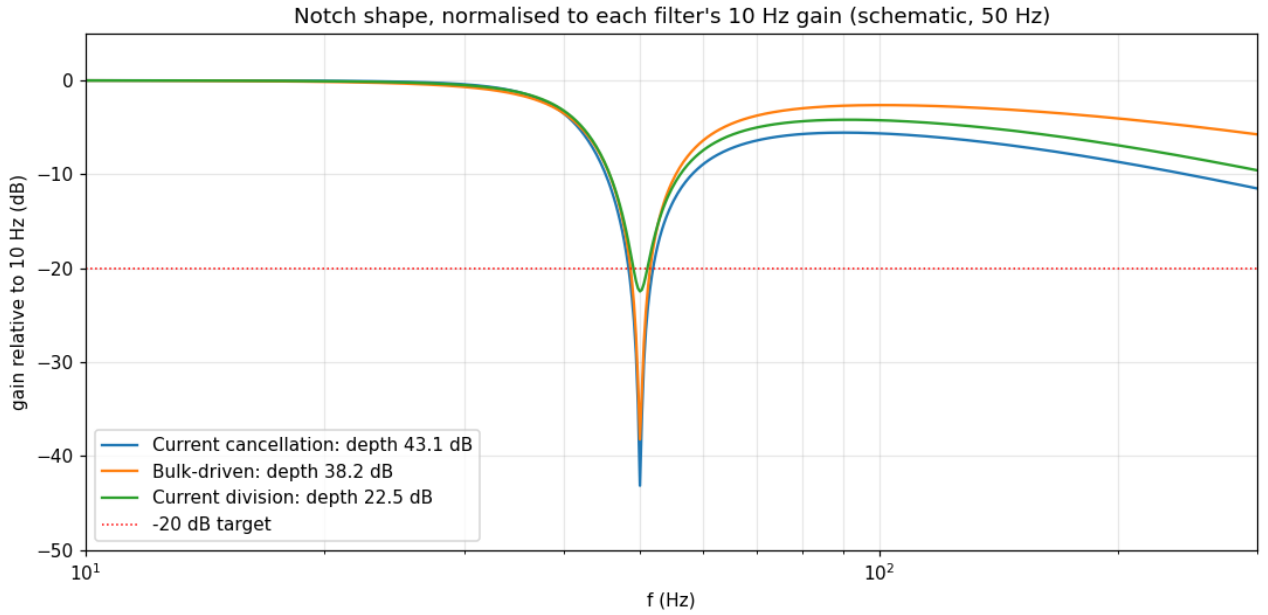


Figure 2. Notch shape with each filter normalised to its own 10 Hz gain.

Normalised, the notch widths are similar, because the capacitor ratios set them. Bulk-driven recovers best above the notch; current cancellation stays lowest, consistent with its large mirror and output capacitance.

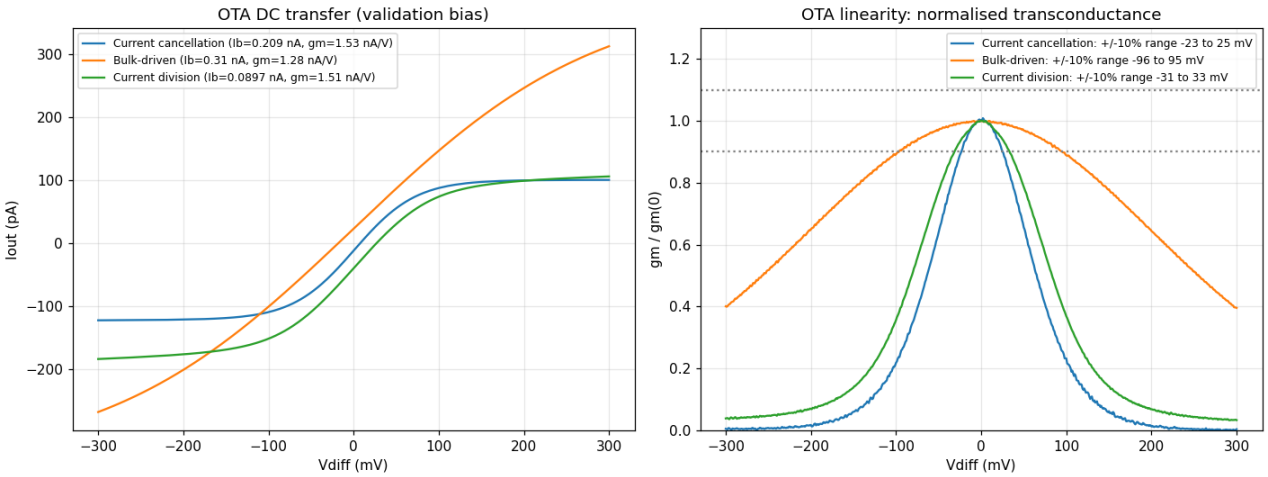


Figure 3. OTA DC transfer and normalised transconductance.

This is the clearest difference between the OTAs. Driving the body divides g_m by about 3, and widens the linear range by the same factor, to about ±95 mV. The two gate-driven styles stay within ±10% of g_m only over about ±25–30 mV.

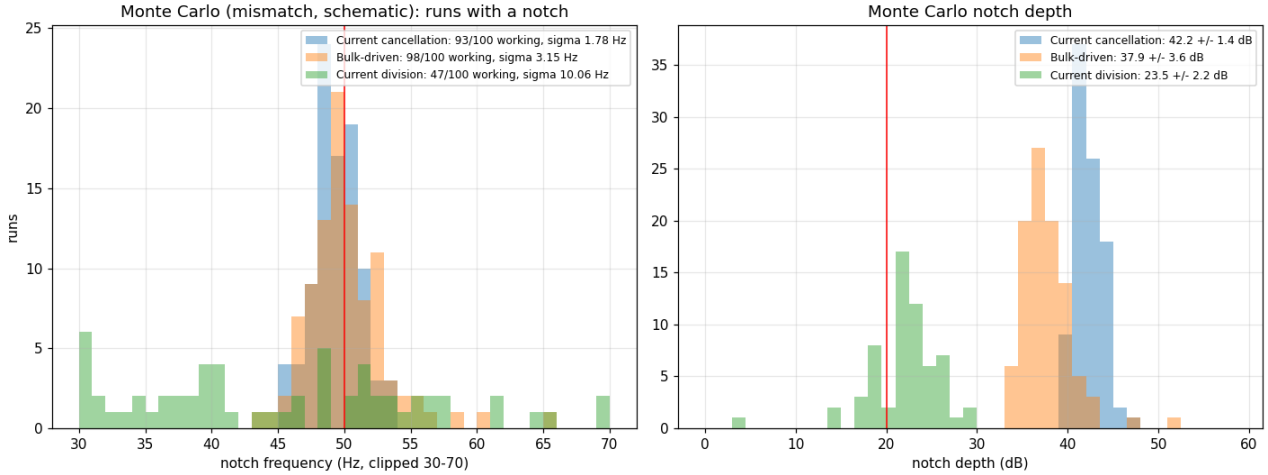


Figure 4. Monte Carlo notch frequency and depth, 100 mismatch runs per style (schematic).

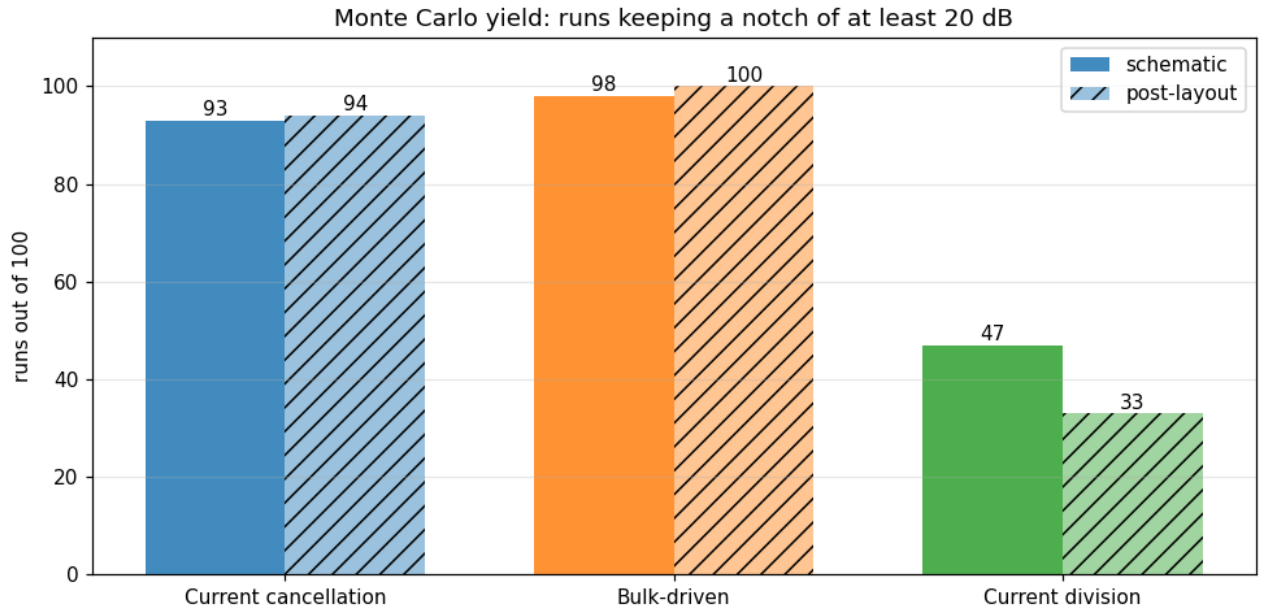


Figure 5. Monte Carlo yield, schematic and post-layout.

Yield follows linear range divided by offset. Bulk-driven keeps nearly every run working. Current cancellation loses about 7% of runs when offsets push an OTA past its narrow range. Current division loses about half, because its systematic offset alone already uses most of its range. Bulk-driven pays for its robustness with a wider f_0 spread (σ about 3 Hz against about 1.8 Hz): the same device mismatch, seen through the weaker body control, moves g_m further.

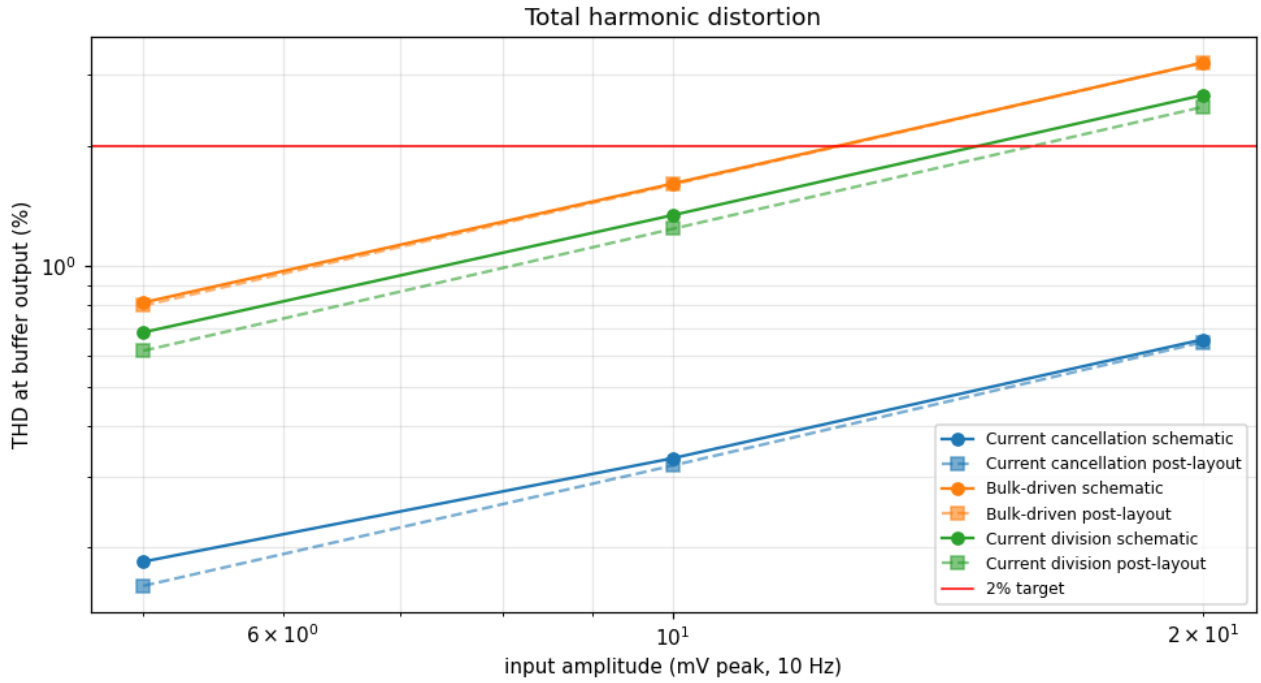


Figure 6. THD at the buffer output vs input amplitude.

Current cancellation has the lowest distortion. The other two are limited by the shared buffer rather than the filter: the bulk-driven filter alone measures 0.04–0.07% THD before the buffer, but its output DC level sits on the non-linear part of the buffer's transfer curve.

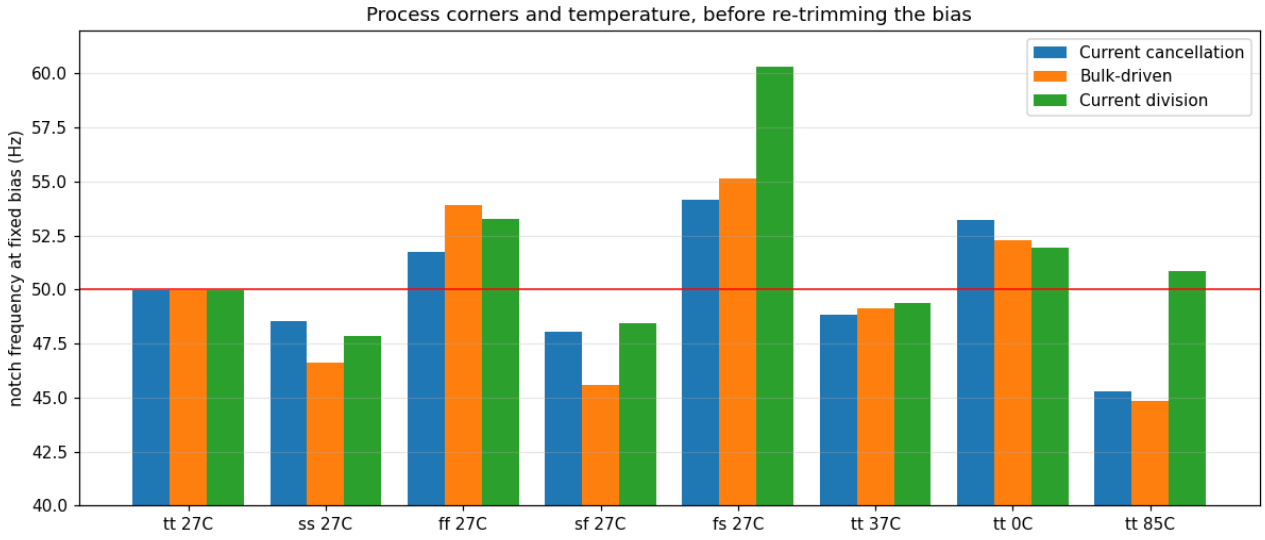


Figure 7. Notch frequency over process corners and temperature at fixed bias.

All three drift by roughly ±5–10 Hz over corners and 0–85 °C, and in each case one bias trim brings f_0 back near 50 Hz. Any of them would need a per-chip or per-temperature trim in practice.

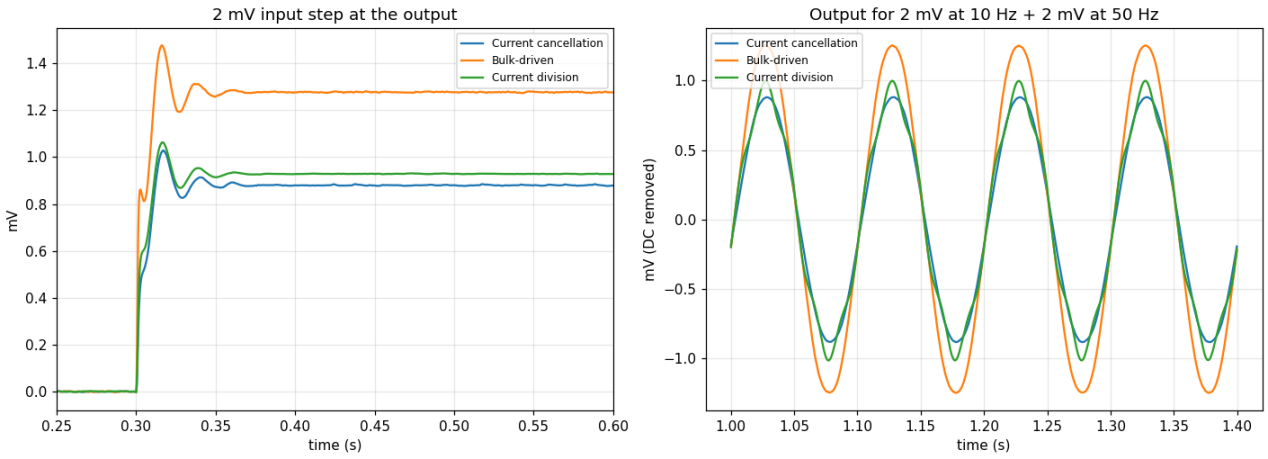


Figure 8. Step response and the 10 Hz + 50 Hz mains-rejection test.

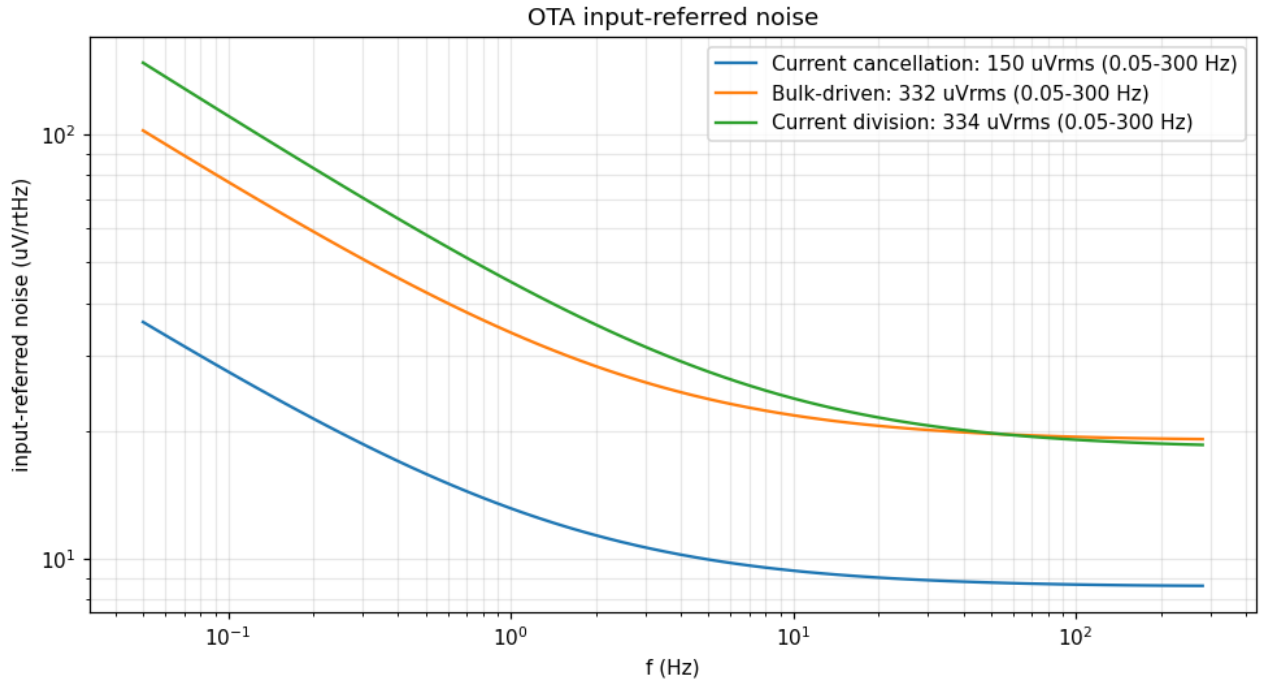


Figure 9. OTA input-referred noise.

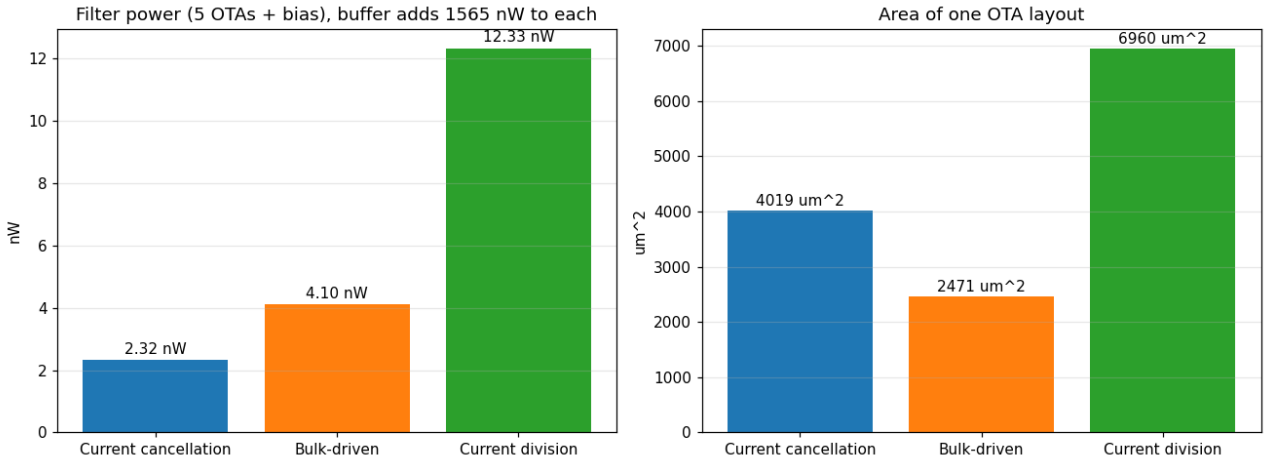


Figure 10. Filter power and single-OTA layout area.

3. Conclusions

Bulk-driven is the most robust choice, and the one I recommend for a manufacturable 50/60 Hz notch in Sky130. It has the best mismatch yield (98–100%), a four-times-wider linear range, the highest CMRR, the smallest layout, and it needed one design cycle. Its costs are about twice the noise, a wider f_0 spread that a bias trim must absorb, about 4 dB of depth lost to layout (still 34 dB), and THD above 2% for inputs over about 12 mV, which comes from the shared buffer, not the OTA.

Current cancellation is the best performer at nominal: the deepest notch, the lowest noise, distortion and power, and the smallest layout penalty. Its 93% yield is acceptable but not comfortable, because a 7.6 mV systematic offset sits against a ±25 mV linear range. It is a good choice if offset trimming is available.

Current division is not recommended in Sky130 as implemented. It meets the targets at nominal, but with about 2.5 dB of depth margin, and fewer than half of the Monte Carlo runs keep a working notch. It also uses the most power and area. Raising its gain with a cascoded mirror would be the way to rescue it, but that departs from the paper's circuit.

4. Limits that apply to all three

• Process variation is covered by corners, because the PDK's process Monte Carlo section fails with a model-bin error; mismatch Monte Carlo uses the tt_mm statistics.

• Bias currents come from ideal sources; an on-chip sub-nanoamp reference and its own spread are not included.

• Only the OTA is laid out; the buffer, capacitors and the wiring between the five OTAs are schematic-level, and wire resistance was not extracted.

• The shared buffer dominates power (1.565 µW) and limits THD for two of the styles; improving it would help all three.

• Sky130 model accuracy at picoamp device currents was not independently checked.